Candidate Number: 295531

1a.

$$Br = \begin{bmatrix} 2 \\ 15 \\ 18 \\ 12 \end{bmatrix} \quad A(Br) = \begin{bmatrix} 345 \\ 200 \\ 118 \\ 111 \end{bmatrix} \quad Cr = \begin{bmatrix} 345 \\ 200 \\ 118 \\ 111 \end{bmatrix}$$

Since A(Br) = Cr return yes

1b.

No as Freivald's algorithm never produces false negatives but can produce false positives. The fact that A(Br) = Cr brings it closer to being true that A · B = C but not certainly true. Currently it is a 1 / 2^k chance of it being true which translates to ½ as we only ran it one time.

1c.

1 / 2^k < 5/100

1 / 2^k < 1/20

2^k > 20

2^4 = 16

2^5 = 32

32 > 20 therefore k = 5

2a.

X(6) = X(n) = 0 as this is a base case

X(5) = X(n-1) = max {Vn-1 ,0}

max {-2,0} = 0 as this is a base case

$X(4) = \max\{X(5), 1 + X(5), 1 \cdot -2 + X(6)\}$

$\max\{0, 1, -2\} = 1$ $X(4) = 1$

$X(3) = \max\{X(4), 2+X(4), 2\cdot 1 + X(5)\}$

$\max\{1, 3, 2\} = 3$ $X(3) = 3$

$X(2) = \max\{X(3), 3+X(3), 3\cdot 2 + X(4)\}$

$\max\{3, 6, 7\} = 7$ $X(2) = 7$

$X(1) = \max\{X(2), -3+X(2), -3\cdot 3 + X(3)\}$

$\max\{7, 4, -6)$ $X(1) = 7$

$X(0) = \max\{X(1), -2 + X(1), -2\cdot -3 + X(2)\}$

$\max\{7, 5, 13\}$ $X(0) = 13$

<u>2b.</u>

X(6) = 0/ none as no pins left

X(5) = v5 considered, best to ignore v5 so

v5

X(4) = v4, v5 considered, best to hit pin v4 and follow X(5) strategy so ignore v5.

<u>v4</u>, v5

X(3) = v3, v4, v5 considered, best to hit v3 and follow X(4) strategy so hit v4 alone and ignore v5.

<u>v3</u>, <u>v4</u>, v5

X(2) = v2, v3, v4, v5 considered, best to hit v2 and v3 together and follow X(4) strategy so hit v4 alone and ignore v5.

v2, v3, v4, v5

X(1) = v1, v2, v3, v4, v5 considered, best to ignore v1 and follow X(2) strategy so hit v2 and v3 together, hit v4 alone and ignore v5.

v1, v2, v3, v4, v5

X(0) = v0, v1, v2, v3, v4, v5 considered, best to hit v0 and v1 together and follow the X(2) strategy so hit v2 and v3 together, then hit v4 alone and ignore v5.

v0, v1, v2, v3, v4, v5

3.

A1	n^2 / 6017	=	n^2
A2	Sqrt(n^3) + n	=	n^3/2
A3	2^(n+3)	=	2^n
A4	n log(n^2)	=	nlog(n)
A5	2025n	=	n
A6	2^n	=	2^n

A5, A4, A2, A1, (A3, A6)

4a.

When the input is A = [0,1,2,3,4,5,6,7,8]

n = 9 >7 so runs lines 3 and onwards

q = 3

Every element is <= 0 so runs line 5.

return mystery(A[: q −1])+mystery(A[q : 2q −1])

For mystery(A[: q −1]) its mystery(0,1,2)

In this case n = 3 <7 so returns n which is 3

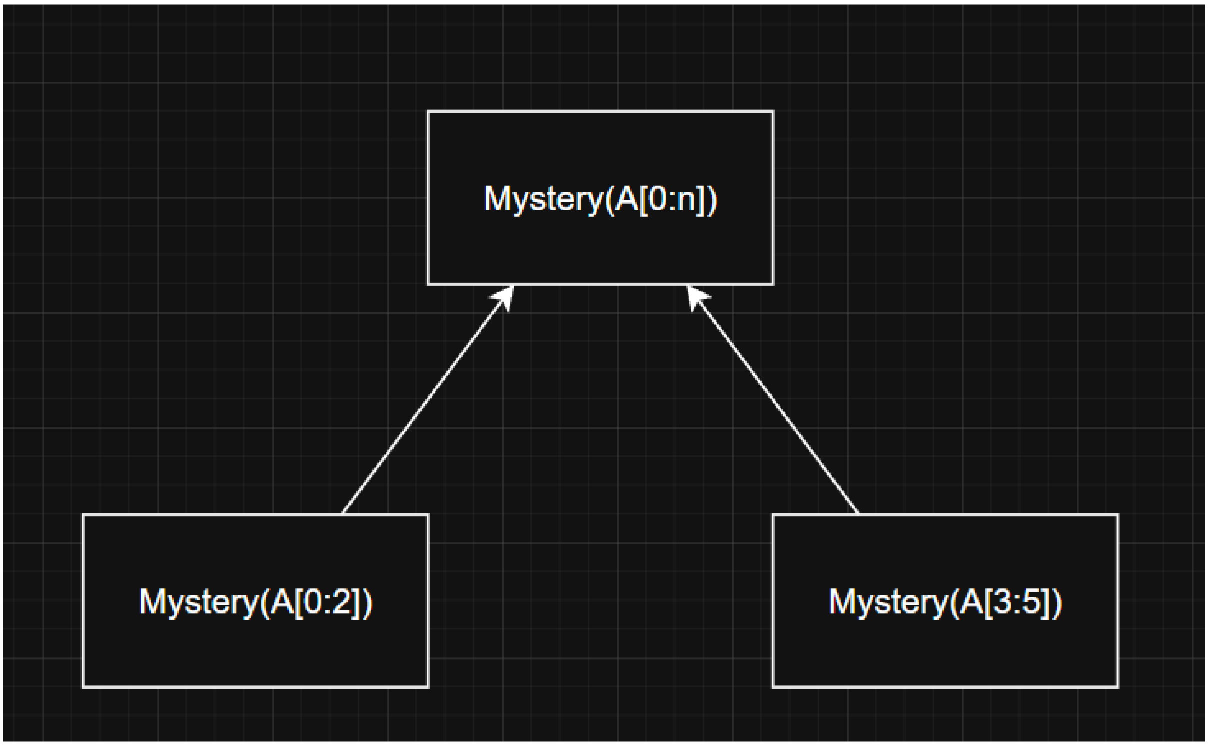
For mystery(A[q : 2q −1]) its mystery(3,4,5)

In this case n = 3 <7 so returns n which is 3

This means this line of code does return 3+3 = 6

Returns 6

4b.



4c.

$T(n) = 2T(n/3) + O(1)$

2T as it runs 2 recursions each time (in worst case)

n/3 as it splits the input into thirds.

O(1) as it takes constant time for non recursive step.

4d.

Master Theorem

a = 2 , b = 3, k = 0

K < $\log_3(2)$ so

T(n) = θ(n^ $\log_3(2)$)

5a.

T(n) = 4T (n/5) +n^2

a = 4, b = 5, k = 2

Height = $\log_b n$ so

Height = $\log_5(n)$

5b.

Level 0, nodes = 1

Level 1, nodes = 4

Level 2, nodes = 16 = 4^2

So at level i, there are 4^i nodes

Node amount = 4^i where i = level

5c.

Size of problem = $\frac{n}{5^i}$ at level i

$\left(\frac{n}{5^i}\right)^2$ as non recursive is n^2 = $\left(\frac{n^2}{5^{2i}}\right)$

Cost at i = nodes · work per node = $4^i \cdot \left(\frac{n^2}{5^{2i}}\right) = n^2 \left(\frac{4}{25}\right)^i$

5d.

Total cost = sum of all levels

So T(n) = $\sum_{i=0}^{\log 5(n)} n^2 \left(\frac{4}{25}\right)^i$

$\left(\frac{4}{25}\right)^i$ is a series so use formula:

$\sum_{i=0}^{k} r^i \;=\; \frac{1-r^{k+1}}{1-r}$

So T(n) = n^2 $\cdot \frac{1-r^{k+1}}{1-r}$

T(n) = n^2 $\cdot \frac{1-\left(\frac{4}{25}\right)^{\log 5(n)+1}}{1-\left(\frac{4}{25}\right)}$

As n grows, 4/25 gets smaller and smaller since 4/25 <1.

Therefore it eventually reaches 0 and can get taken out of the equation.

So T(n) = n^2 $\cdot \frac{1}{1-\left(\frac{4}{25}\right)}$

Since $\frac{1}{1-\left(\frac{4}{25}\right)}$ is constant it gets taken away leaving T(n) = n^2

So T(n) = θ(n^2)

Master theorem to double check:

a = 4, b = 5, k = 2

2 > $\log_5 4$ so T(n) = n^k = n^2

So T(n) = θ(n^2) which shows its correct

<u>6a.</u>

T(n) = 7T (n/3) +√n

a = 7, b = 3, k = ½

½ < $\log_3 7$ so T(n) = θ(n^ $\log_3 7$)

6b.

$T(n) = 16T(n/4) + n^2$

$a = 16, b = 4, k = 2$

$2 = \log_4 16$ so $T(n) = \theta(n^2 \log(n))$

6c.

$T(n) = T(n/2) + 7$

$a = 1, b = 2, k = 0$

$0 = \log_2 1$ so $T(n) = \theta(n^0 \log(n)) = \theta(\log(n))$

7ai.

$T(n) = 4T(n/4) + \theta(n)$

4T as it runs 4 recursions each time

n/4 as it splits into quarters of the input each time

$\theta(n)$ as the non recursive step runs in $\theta(p+q)$ time where p+q is equal to the size of array X and array Y which in this case adds up to n making it $\theta(n)$

7aii.

Master theorem:

$a = 4, b = 4, k = 1$

$1 = \log_4 4$ so $T(n) = \theta(n^1 \log n) = \theta(n\log(n))$

This is the same as merge sort which makes sense as its using pretty much the exact same algorithm.

7b.

Need to check all the cases

If $a < b$

Then $\log_b a < 1$ meaning that master theorem has a different output.

Master theorem will now be $k > \log_b a$ meaning $T(n) = n^k = n$ since $k = 1$

This can't be correct as the shortest amount of time a sorting algorithm runs in is $\Omega(n\log n)$ and since this is $\theta(n)$ it means it doesn't correctly sort A, must skip out on some steps

If a = b

Then $\log_b a = 1$ meaning master theorem has same output making it the same speed as merge sort.

If a>b

Then $\log_b a > 1$ so different output for master theorem again.

Master theorem will now be $K < \log_b a$ meaning $T(n) = \theta(n^{\log_b a})$

Since $\log_b a > 1$, this makes this a polynomial time complexity which is slower that a linearithmic (nlogn) time complexity meaning this algorithm would be slower.

So if a < b then it isn't a valid sorting algorithm

If a >= b then it is either slower (if a > b) or same speed (a = b) than merge sort.

8a.

SRTBOT

Subproblems: X(i) = max amount her happiness can increase on days $d_i,\ldots,d_{n-1}$.

Relate:

Need a counter for how many days already outside, and then max value so

X(i, j) where i is the max value and j is the amount of days outside

$X(i, j) = \{ 0$ if $i = n$

$\{ \max \{X(i+1,0),\ \ t_i + X(i+1, j+1)$ only if $k < 2\}$ if $i < n$

Where the double { on separate lines is supposed to be one big one like so: 

Topological Order: decreasing values of i

Base Cases: X(n, j) = 0 for any value of j.

Original Subproblem: X(0)

Time Complexity:

Number of subproblems = 3n as there is n amount of days and each day needs to be checked 3 times for the values of j which are 0, 1 and 2. each subproblem takes O(1) time therefore time complexity is $3n \cdot O(1) = \theta(3n) = \theta(n)$

SRTBOT (clean version)

Subproblems: X(i) = max amount her happiness can increase on days $d_i,...,d_{n-1}$.

Relate: X(i, j) = { 0 if i = n

{ max {X(i+1,0), t_i + X(i+1, j+1) only if k <2} if i < n

Topological Order: decreasing values of i

Base Cases: X(n, j) = 0 for any value of j.

Original Subproblem: X(0)

Time Complexity: $O(1) \cdot \theta(n) = \theta(n)$

<u>8bi.</u>

Input T = [2,1,4,2,3]

X(4, 0) = max { X(5,0), 3 +X(5,1) } = max {0,3} = 3

X(4, 1) = max {X(5,0), 3 + X(5,2) } = max {0,3} = 3

X(4, 2) = max{X(5,0)} = max{0} = 0 as j > 2 so cant do t_i + X(i+1, j+1)

X(3, 0) = max {X(4,0), 2+X(4,1)} = max {3, 5} = 5

X(3, 1) = max {X(4,0), 2+X(4,2)} = max {3, 2} = 3

X(3, 2) = max {X(4,0)} = max {3} = 3 as j > 2 so cant do ti + X(i+1, j+1)

X(2, 0) = max {X(3,0), 4 +X(3,1)} = max {5, 7} = 7

X(2, 1) = max {X(3,0), 4 +X(3,2)} = max {5, 7} = 7

X(2, 2) = max {X(3,0)} = max{5} = 5 as j > 2 so cant do ti + X(i+1, j+1)

X(1, 0) = max {X(2,0), 1 + X(2,1)} = max {7, 8} = 8

X(1, 1) = max {X(2,0), 1 + X(2,2)} = max {7, 6} = 7

X(1, 2) = max {X(2,0)} = max{7} = 7 as j > 2 so cant do ti + X(i+1, j+1)

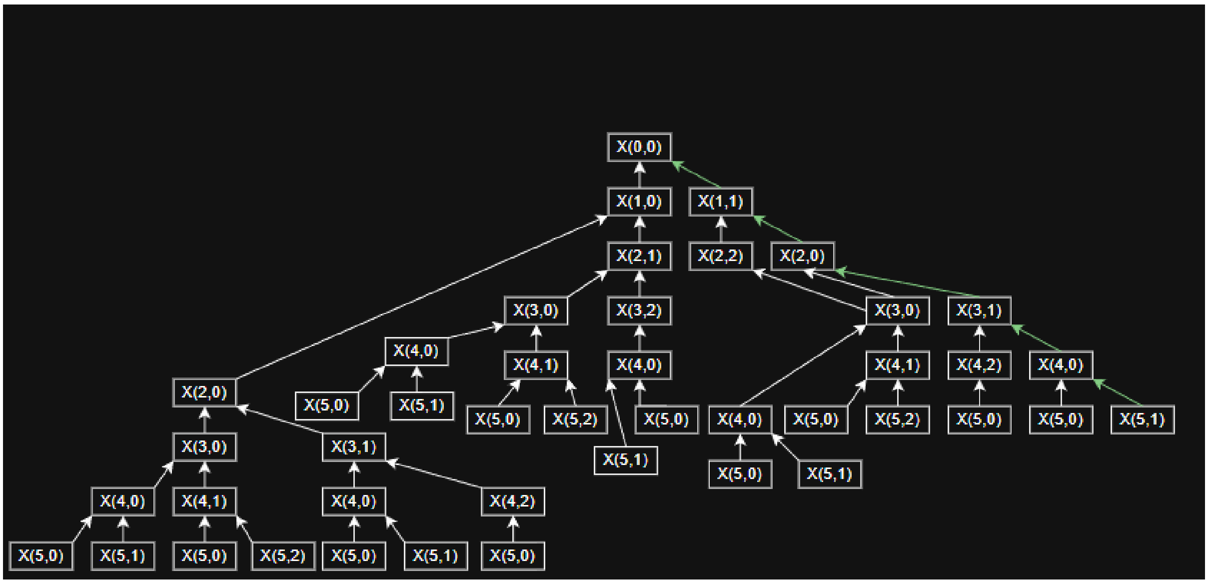
X(0, 0) = max {X(1,0), 2 + X(1,1)} = max {8, 9} = 9

X(0, 1) = max {X(1,0), 2 + X(1,2)} = max {8, 9} = 9

X(0, 2) = max {X(1,0)} = max {8} = 8 as j > 2 so cant do ti + X(i+1, j+1)

So for best output, need to follow:

X(0,0), X(1,1), X(2,0), X(3,1), X(4,0), X(5,1)



8bii.

Start from day n

$X(n,0) = X(n,1) = X(n,2) = 0$ as this is a base case

$X(5,1) = 0$

$X(4,0) = \max\{ X(5,0), 3 + X(5,1) \} = \max\{0,3\} = 3$ so go outside on day 5

$X(3,1) = \max\{X(4,0), 2+X(4,2)\} = \max\{3, 2\} = 3$ so stay inside on day 4

$X(2, 0) = \max\{X(3,0), 4 + X(3,1)\} = \max\{5, 7\} = 7$ so go outside on day 3

$X(1, 1) = \max\{X(2,0), 1 + X(2,2)\} = \max\{7, 6\} = 7$ so stay inside on day 2

$X(0, 0) = \max\{X(1,0), 2 + X(1,1)\} = \max\{8, 9\} = 9$ so go outside on day 1

So in lecture terms,

$[\underline{2},1,\underline{4},2,\underline{3}] = 9$